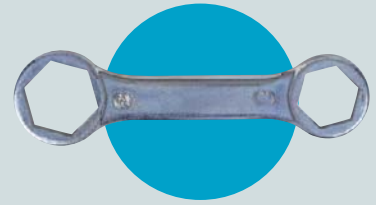




Topology is all about what happens to shapes when they stretch, twist, and tangle. Some people call it “*rubber sheet*” geometry. If you can stretch and bend a shape to make another, the two shapes are “topologically equivalent”. A square and a circle are topologically equivalent because you can stretch one to make the other. But the figures 8 and 0 are not, because the 8 has a connection in the middle.

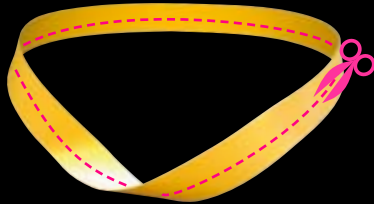
LOOK AT
THESE SHAPES...



... and see if you can tell which of the objects below are *topologically* equivalent to them and which is in a class of its own.



3



Now for something very surprising. What do you think will happen if you cut all the way along the strip in the middle? Try it and see.

4



Make another Möbius strip. This time, cut along it a third of the way from the edge for another surprise.

Now make two circular ribbons of paper (not Möbius strips) and glue them together at one point. What shape do you think they will make if you cut along the middle of both ribbons?



And for my next trick...

Loop a strip of paper or a £10 note into a zigzag and use two paperclips to hold it in shape. Ask a friend what they think will happen if you pull the ends. Give the ends a sharp tug and the paperclips will fly off, linked together! It's magic!



A monkey puzzle

- Early one day, a monk set off to walk to a monastery at the top of a mountain. The path was steep, and it took him all day.
- The next day he returned down the same path, but set off much later and finished the walk in half the time. Was there any point where the monk was in exactly the same place at the same time on both days?